

Cluster Sampling with Equal Probabilities

STA304: Surveys, Sampling, and Observational Data

Emily Somerset, Thursday May 28, 2026

Learning Outcomes

- ❑ Relate the efficiency of a cluster sample to the amount of within-cluster and between-cluster variation
- ❑ Interpret the within-cluster and between-cluster components of variation using an ANOVA decomposition.
- ❑ Explain why cluster sampling is often less precise than a simple random sample of the same size.
- ❑ Use information about within-cluster variation, between-cluster variation, and the ICC to make recommendations about cluster sampling designs.
- ❑ Balance precision and cost considerations when choosing the number of clusters and the number of units sampled within each cluster.

Last class

- We talked about one-stage and two-stage cluster sampling designs.
- We talked about how these cluster sampling designs produce estimators that are in general less efficient than those under an SRS.
 - e.g. $\text{MSE}(\hat{t}_{\text{cluster}}) > \text{MSE}(\hat{t}_{\text{SRS}})$ for a nM observations
 - In one-stage cluster design: These nM observations come from n randomly selected clusters and M not randomly selected units in each sampled cluster (census).
 - Units in the same cluster are generally homogeneous in some important way, often unmeasured way, related to the outcome of interest y
 - So every additional observation within a cluster doesn't offer too much "new" information.
 - This is the intuition behind what is happening.

Last class

- To understand this intuition a bit deeper we will look at an ANOVA table.
- This summarizes data by partitioning total data variability into specific sources: Within-group variability and between-group variability.

5.4 Comparing Clusters with an SRS

Population ANOVA table - cluster sampling

- This is an ANOVA table
- It decomposes the total variance

TABLE 5.1

Population ANOVA Table—Cluster Sampling

Source	df	Sum of Squares	Mean Square
Between psus	$N - 1$	$SSB = \sum_{i=1}^N \sum_{j=1}^M (\bar{y}_{iU} - \bar{y}_U)^2$	MSB
Within psus	$N(M - 1)$	$SSW = \sum_{i=1}^N \sum_{j=1}^M (y_{ij} - \bar{y}_{iU})^2$	MSW
Total, about $\bar{y}_U$	$NM - 1$	$SSTO = \sum_{i=1}^N \sum_{j=1}^M (y_{ij} - \bar{y}_U)^2$	S^2

5.4 Comparing Clusters with an SRS

Population ANOVA table - cluster sampling

- This is an ANOVA table
- It decomposes the total variance
- Into the sum of the within-group and between-group variance.

TABLE 5.1

Population ANOVA Table—Cluster Sampling

Source	df	Sum of Squares	Mean Square
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5.4 Comparing Clusters with an SRS

Population ANOVA table - cluster sampling

- **Stop and think:** Take a look at each of these formulas and explain in your own words what it measures.

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5.4 Comparing Clusters with an SRS

Population ANOVA table - cluster sampling

- The degree of information measures how many **independent** pieces of information you have.

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5.4 Comparing Clusters with an SRS

Population ANOVA table - cluster sampling

- Example:

$$SSB = \sum_{i=1}^N \sum_{j=1}^M (\bar{y}_{i\mathcal{U}} - \bar{y}_{\mathcal{U}})^2$$

5.4 Comparing Clusters with an SRS

Population ANOVA table - cluster sampling

- The mean square column is the sum of squares column divided by the degrees of freedom.

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5.4 Comparing Clusters with an SRS

Connection to the total estimator

- The variance of the total estimator was:

$$V(\hat{t}) = N^2 \frac{S_t^2}{n} \left(1 - \frac{n}{N} \right)$$

- And depends on S_t^2

$$S_t^2 = \frac{1}{N-1} \sum_{i=1}^N (t_i - t/N)^2 =$$

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Population ANOVA Table—Cluster Sampling

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5.4 Comparing Clusters with an SRS

Connection to the total estimator

- The variance of the total estimator was:

•
$$V(\hat{t}) = N^2 \frac{S_t^2}{n} \left(1 - \frac{n}{N} \right)$$

- And so in terms of MSB:
- $V(\hat{t}) =$

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5.4 Comparing Clusters with an SRS

Connection to the total estimator

- $V(\hat{t}) = \frac{N^2 M}{n} \left(1 - \frac{n}{N} \right) \times \text{MSB}$
- Note on this intuitively:

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Population ANOVA Table—Cluster Sampling

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5.4 Comparing Clusters with an SRS

Revisiting comparing precision with an SRS of the same size

- 1-stage cluster nM observations:

- $$V(\hat{t}_{\text{cluster}}) = \frac{N^2 M}{n} \left(1 - \frac{n}{N} \right) \times \text{MSB}$$

- SRS with nM observations:

- $$V(\hat{t}_{\text{SRS}}) = (NM)^2 \frac{S^2}{nM} \left(1 - \frac{nM}{NM} \right)$$

TABLE 5.1

Population ANOVA Table—Cluster Sampling

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Revisiting comparing precision with an SRS of the same size

- 1-stage cluster nM observations:

- $$V(\hat{t}_{\text{cluster}}) = \frac{N^2 M}{n} \left(1 - \frac{n}{N} \right) \times \text{MSB}$$

- SRS with nM observations:

- $$V(\hat{t}_{\text{SRS}}) = (NM)^2 \frac{S^2}{nM} \left(1 - \frac{nM}{NM} \right)$$

- Relative efficiency:

- $$\frac{V(\hat{t}_{\text{cluster}})}{V(\hat{t}_{\text{SRS}})} =$$

5.4 Comparing Clusters with an SRS

Revisiting comparing precision with an SRS of the same size

- Let's relate $\frac{V(\hat{t}_{\text{cluster}})}{V(\hat{t}_{\text{SRS}})} = \frac{\text{MSB}}{S^2}$ to the intraclass coefficient ICC.
- Recall from last class: The ICC provides a measure for **homogeneity within clusters**
- It's defined as the Pearson correlation coefficient for the $NM(M - 1)$ pairs of (y_{ij}, y_{ik}) observations for $i = 1, \dots, N, j \neq k$

$$\text{ICC} = \frac{\sum_{\text{pairs}} (y_{ij} - \bar{y}_{\mathcal{U}})(y_{ik} - \bar{y}_{\mathcal{U}})}{\sum_{\text{pairs}} (y_{ij} - \bar{y}_{\mathcal{U}})^2}$$

Measures the correlation between all pairs from a same cluster across the population.

5.4 Comparing Clusters with an SRS

Revisiting comparing precision with an SRS of the same size

- It can be shown that:

$$ICC = \frac{\sum_{\text{pairs}} (y_{ij} - \bar{y}_{\mathcal{U}})(y_{ik} - \bar{y}_{\mathcal{U}})}{\sum_{\text{pairs}} (y_{ij} - \bar{y}_{\mathcal{U}})^2} = 1 - \frac{M}{M - 1} \frac{SSW}{SSTO}$$

$$0 \leq \frac{SSW}{SSTO} \leq 1 \qquad -\frac{1}{M - 1} \leq ICC \leq 1$$

TABLE 5.1 Population ANOVA Table—Cluster Sampling			
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5.4 Comparing Clusters with an SRS

Revisiting comparing precision with an SRS of the same size

$$ICC = \frac{\sum_{\text{pairs}} (y_{ij} - \bar{y}_{\mathcal{U}})(y_{ik} - \bar{y}_{\mathcal{U}})}{\sum_{\text{pairs}} (y_{ij} - \bar{y}_{\mathcal{U}})^2} = 1 - \frac{M}{M-1} \frac{SSW}{SSTO}$$

- As within-cluster variation increases, clusters are more _____,
and ICC approaches _____
- As within cluster variation decreases, clusters are more _____,
and ICC approaches _____

5.4 Comparing Clusters with an SRS

Revisiting comparing precision with an SRS of the same size

- ICC < 0 when $\frac{SSW}{SSTO} > \frac{M - 1}{M}$
- If the Pearson correlation is negative, within each cluster, high and low values are systematically paired together.
- A randomly assigned group of people into each cluster would just reflect population variance.
- But if clustering actively mixes opposites together, the within spread exceeds what you would get by chance -> **more heterogeneous than the population.**

5.4 Comparing Clusters with an SRS

Revisiting comparing precision with an SRS of the same size

Example: consider two populations each with 3 psu and 3 ssu in each psu:

	Population A			Population B				Population A		Population B	
								$\bar{y}_{iU}$	S_i^2	$\bar{y}_{iU}$	S_i^2
psu 1	10	20	30	9	10	11	psu 1	20	100	10	1
psu 2	11	20	32	17	20	20	psu 2	21	111	19	3
psu 3	9	17	31	31	32	30	psu 3	19	124	31	1

- Both populations have the same elements (numbers) just organized differently
- So they share $\bar{y}_{\mathcal{U}} = 20$ and $S^2 = 84.5$

5.4 Comparing Clusters with an SRS

Revisiting comparing precision with an SRS of the same size

Example: consider two populations each with 3 psu and 3 ssu in each psu:

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psu 2	11	20	32	17	20	20	psu 2	21	111	19	3
psu 3	9	17	31	31	32	30	psu 3	19	124	31	1

- In Population A, the means of each cluster are similar and there's lots of within-cluster variability (S_i^2)
- In Population B, there's less within-cluster variability (S_i^2) but more variation between the cluster means.

5.4 Comparing Clusters with an SRS

Revisiting comparing precision with an SRS of the same size

Example: consider two populations each with 3 psu and 3 ssu in each psu:

ANOVA Table for Population A:

Source	df	SS	MS
Between psus	2	6	3
Within psus	6	670	111.67
Total, about mean	8	676	84.5

$$ICC = 1 - \left(\frac{3}{2}\right) \frac{670}{676} = -0.4867$$

ANOVA Table for Population B:

Source	df	SS	MS
Between psus	2	666	333
Within psus	6	10	1.67
Total, about mean	8	676	84.5

$$ICC = 1 - \left(\frac{3}{2}\right) \frac{10}{676} = 0.9778$$

- We can see that the ICC in population A is negative and positive in Population B.

5.4 Comparing Clusters with an SRS

Revisiting comparing precision with an SRS of the same size

	Population A			Population B				Population A		Population B	
								$\bar{y}_{iU}$	S_i^2	$\bar{y}_{iU}$	S_i^2
psu 1	10	20	30	9	10	11	psu 1	20	100	10	1
psu 2	11	20	32	17	20	20	psu 2	21	111	19	3
psu 3	9	17	31	31	32	30	psu 3	19	124	31	1

- In population A, every cluster has a small, medium, and large value.
- So if you pick two elements from the same cluster, you're guaranteed to get one low one high (negatively correlated).
- This is more dispersed than what you'd expect just drawing from the entire population.

5.4 Comparing Clusters with an SRS

Revisiting comparing precision with an SRS of the same size

$$\frac{V(\hat{t}_{\text{cluster}})}{V(\hat{t}_{\text{SRS}})} = \frac{NM - 1}{M(N - 1)} [1 + (M - 1)\text{ICC}]$$

Summary:

- Only when ICC is negative, would cluster estimators be more efficient than SRS estimators.
- If ICC is positive, PSUs in the same cluster tend to be similar, and cluster sampling is less efficient than simple random sampling of elements.

5.5 Designing a Cluster Sample

Design Questions

1. What overall precision is needed?
2. What size should the psus be? (AKA how big should M be?)
3. How many ssus should be sampled in each psu selected for the sample?
4. How many psus should be sampled?

Q1 is faced in any survey design. Often a conversation with experts.

5.5 Designing a Cluster Sample

Size of the psus

- Often psus are naturally forming and this decision is made for us.
 - E.g. classrooms, schools, farms, egg cartons
- This means our decision on the size of M would be made.
- Can be more difficult like splitting up a city into geographical areas.
- How large should each psu be? 1km^2 , 10km^2 , 20km^2 ?

5.5 Designing a Cluster Sample

Size of the psus

- General principle in area surveys is that the larger the psu the better because the more variability you are likely to see in a psu.
- However if the psu size is too large, you might lose the cost savings of cluster sampling.

5.5 Designing a Cluster Sample

Choosing subsampling sizes

- Suppose we have chosen a psu size
- Let $M_i = M$ and $m_i = m$ for all psus (simplest case)
- Then:

$$V(\hat{\bar{y}}) = \left(1 - \frac{n}{N}\right) \frac{\text{MSB}}{nM} + \left(1 - \frac{m}{M}\right) \frac{\text{MSW}}{nm}$$

5.5 Designing a Cluster Sample

Choosing subsampling sizes

- Suppose we have chosen a psu size
- Let $M_i = M$ and $m_i = m$ for all psus.
- Then:

$$V(\hat{y}) = \left(1 - \frac{n}{N}\right) \frac{\text{MSB}}{nM} + \left(1 - \frac{m}{M}\right) \frac{\text{MSW}}{nm}$$

If $\text{MSW} = 0$ (no within-variation), you may as well set $m = 1$. Adding more subsamples doesn't decrease precision of the estimator.

5.5 Designing a Cluster Sample

Choosing subsampling sizes

$$V(\hat{y}) = \left(1 - \frac{n}{N}\right) \frac{\text{MSB}}{nM} + \left(1 - \frac{m}{M}\right) \frac{\text{MSW}}{nm}$$

Often you'll combine this variance formula with a cost function:

$$\text{Total cost} = C = n(c_1 + c_2 m)$$

Example: $c_1 = \$30$ = cost to visit a neighbourhood (cluster)

$c_2 = \$10$ = cost to visit each house (ssu) in neighbourhood.

5.5 Designing a Cluster Sample

Choosing subsampling sizes

$$V(\hat{y}) = \left(1 - \frac{n}{N}\right) \frac{\text{MSB}}{nM} + \left(1 - \frac{m}{M}\right) \frac{\text{MSW}}{nm}$$

Often you'll combine this variance formula with a cost function:

$$\text{Total cost} = C = n(c_1 + c_2m)$$

Example: $c_1 = 40$ minutes per dorm room

$c_2 = 10$ minutes per student.

5.5 Designing a Cluster Sample

Choosing subsampling sizes

$$V(\hat{y}) = \left(1 - \frac{n}{N}\right) \frac{\text{MSB}}{nM} + \left(1 - \frac{m}{M}\right) \frac{\text{MSW}}{nm}$$

Often you'll combine this variance formula with a cost function:

$$\text{Total cost} = C = n(c_1 + c_2m)$$

Now you have an optimization problem! Minimize V constrained to this cost function, for a fixed total cost C

5.5 Designing a Cluster Sample

Choosing subsampling sizes

The textbook actually has an exercise to solve this optimization problem. The answer is:

$$n_{\text{opt}} = \frac{C}{c_1 + c_2 m_{\text{opt}}}$$

$$m_{\text{opt}} = \sqrt{\frac{c_1 M (N - 1) (1 - R_a^2)}{c_2 (NM - 1) R_a^2}}$$

$$\text{Here } R_a^2 = 1 - \frac{\text{MSW}}{S^2}$$

Another measure
of within-group
homogeneity
(adjusted R^2)

If you didn't have this population quantity you could estimate it using past survey data or other observational data.

$$\hat{R}_a = 1 - \frac{\hat{\text{MSW}}}{\hat{S}^2} \text{ (from ANOVA table)}$$

5.5 Designing a Cluster Sample

Choosing subsampling sizes

$$V(\hat{\bar{y}}) = \left(1 - \frac{n}{N}\right) \frac{\text{MSB}}{nM} + \left(1 - \frac{m}{M}\right) \frac{\text{MSW}}{nm}$$

Minimize this variance formula with respect to a cost function:

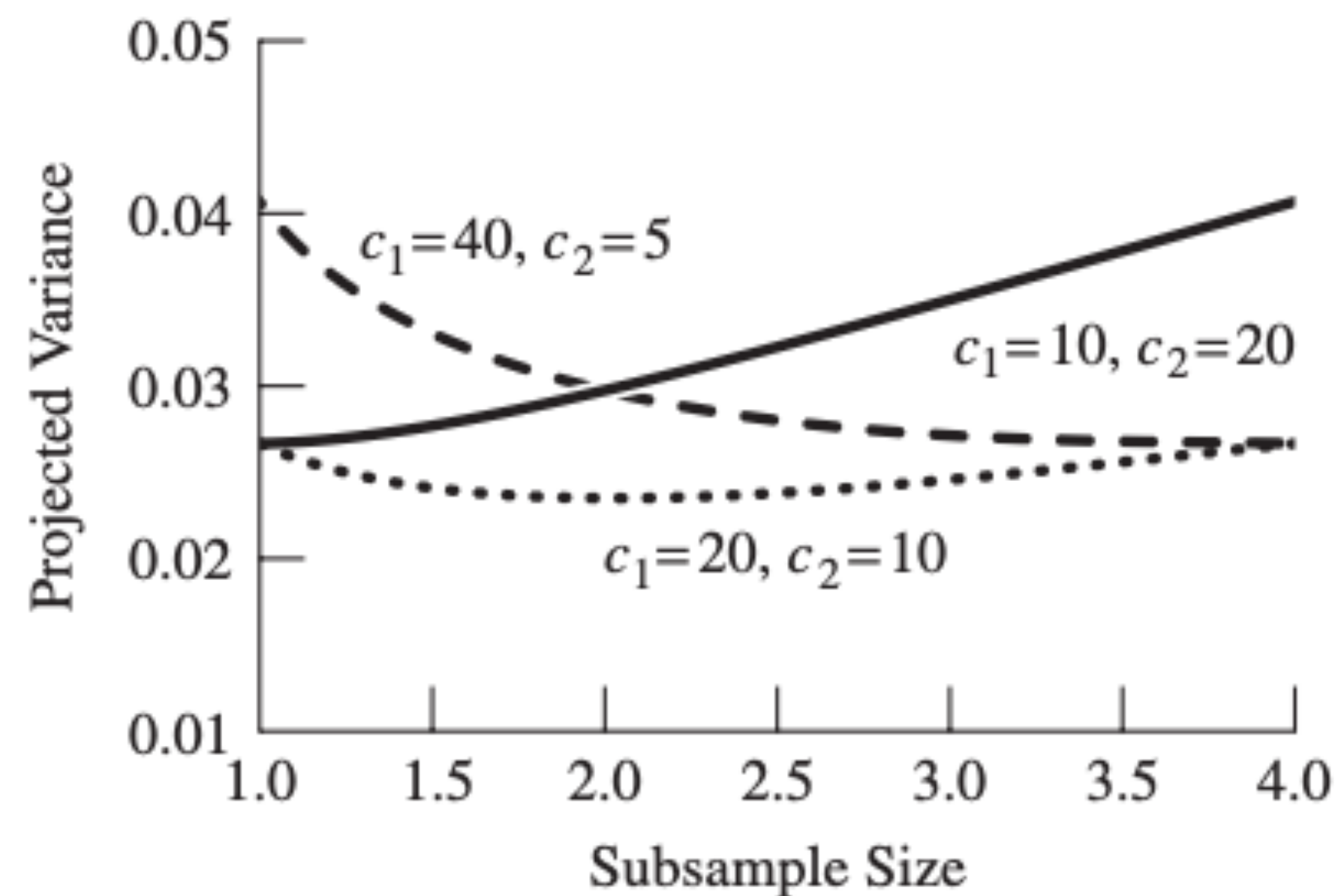
$$\text{Total cost} = C = n(c_1 + c_2m)$$

Or you can plot the variance versus the cost for a variety of subsamples m and n .

5.5 Designing a Cluster Sample

Choosing subsampling sizes

Or you can plot the variance versus the cost for a variety of subsamples m .



- Here $\hat{R}_\alpha$ was set based on past data
- C was fixed.
- c_1, c_2, m were varied and plotted against variance

5.5 Designing a Cluster Sample

Choosing subsampling sizes

- Summary:
- For design purposes we only need an estimate of ICC or R^2_α or MSW and MSB

We obtain the following ANOVA table for the coots data in Example 5.7.

Source	DF	Sum of Squares	Mean Square	F Value
Model	183	257.4175336	1.4066532	237.44
Error	184	1.0900782	0.0059243	
Corrected Total	367	258.5076118		
R-Square	C.V.	Root MSE	VOLUME Mean	
0.995783	3.298616	0.076970	2.333394	

High degree of homogeneity within groups ($R^2_\alpha \approx 0.996$)

5.5 Designing a Cluster Sample

Choosing the sample size (number of psus)

- Once you determine M and m and you have a desired precision, you can determine n usual methods.
- That is, construct the theoretical half width of a confidence interval and set it to the desired precision and solve for n !
- You can use past data to estimate parameters that may be needed (e.g. s_i^2, s_t^2).

$$\hat{V}(\hat{t}) = N^2 \left(1 - \frac{n}{N}\right) \frac{s_t^2}{n} + \frac{N}{n} \sum_{i \in S} \left(1 - \frac{m_i}{M_i}\right) M_i^2 \frac{s_i^2}{m_i} \quad z_{\alpha/2} \sqrt{\hat{V}(\hat{t})} = \text{Desired Precision}$$

5.5 Designing a Cluster Sample

Summary

- All these choices of M , n , and m might need to be reiterated until you find a combination that works.
- Increasing M to increase heterogeneity.
- And then figuring out an n and an m that will give a low variance with a desired level of precision.

Learning activity 8 will briefly touch on these ideas.

Progress

- ☑ Relate the efficiency of a cluster sample to the amount of within-cluster and between-cluster variation
- ☑ Interpret the within-cluster and between-cluster components of variation using an ANOVA decomposition.
- ☑ Explain why cluster sampling is often less precise than a simple random sample of the same size.
- ☑ Use information about within-cluster variation, between-cluster variation, and the ICC to make recommendations about cluster sampling designs.
- ☑ Balance precision and cost considerations when choosing the number of clusters and the number of units sampled within each cluster.